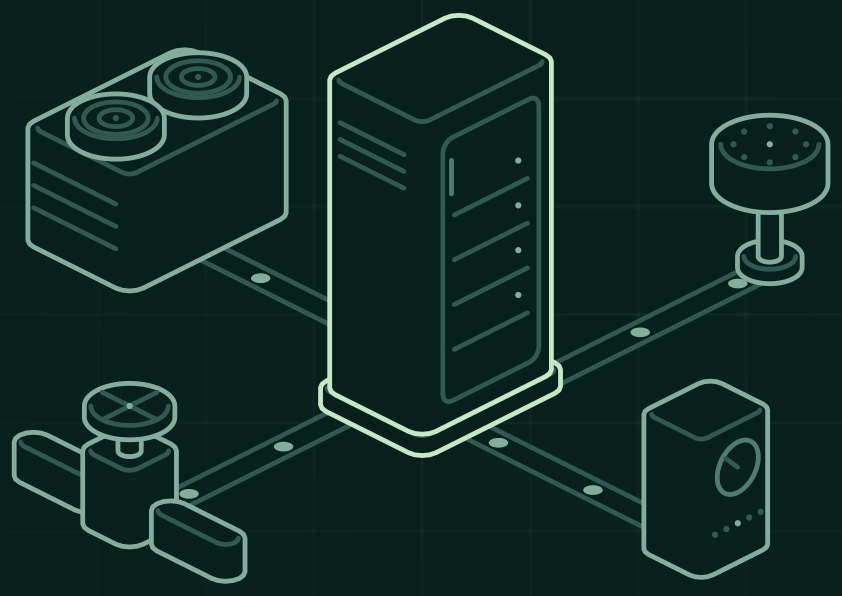
• OWNERS · FACILITY MANAGERS · TECHNICAL DIRECTORS

You don’t buy a BMS. You specify it.

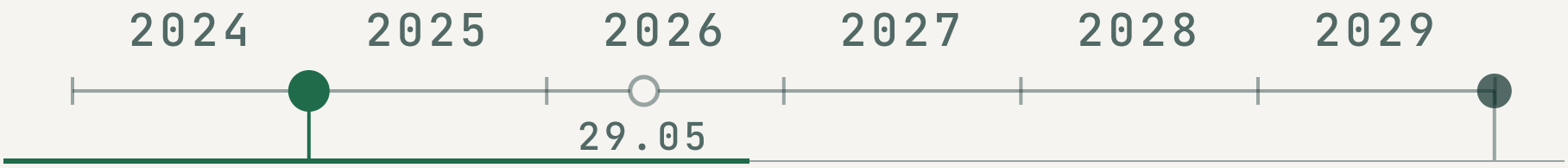
Romania’s BACS obligation for non-residential buildings with over 290 kW of heating or air conditioning has been due since 31.12.2024. What the law requires, what it costs, and what to write into the specification so you are not tied to one supplier.

FIG. 01 · ONE BMS, FOUR SOURCES



• LEGAL OBLIGATION

Two thresholds.
Two legal regimes.



31.12.2024

290 kW

Law 372/2005 (Romania)

Deadline passed: the obligation already applies.

Non-residential buildings, effective rated output per system family (heating, air conditioning or combined with ventilation), where technically and economically feasible.

31.12.2029

70 kW

Directive (EU) 2024/1275

Not yet transposed into Romanian law: not a domestic obligation today.

Transposition deadline: 29.05.2026. On 15.07.2026 the Commission sent letters of formal notice to all 27 member states.

Law 238/2024 raised the fines under Law 372/2005 by up to about 400%.

• DEFINITION

It measures. It controls. It records.

A BMS is the layer of control and measurement between a building's services (heating, ventilation, air conditioning, lighting, pumps, metering) and the people responsible for them.

> Without it, each piece of plant runs in isolation: the boiler only knows the boiler, the chiller only knows the chiller, and nobody knows whether both are running at once on an April day.

In law it is BACS, building automation and control systems: the same technical object. Not to be confused with the other BMS, battery management system.

• ARCHITECTURE

Three levels.

Each with its own cost risk.

L3 Supervision

Server, operator station (HMI), licences.

Licences are the most often forgotten budget line.

L2 Automation

DDC controllers and the automation panel.

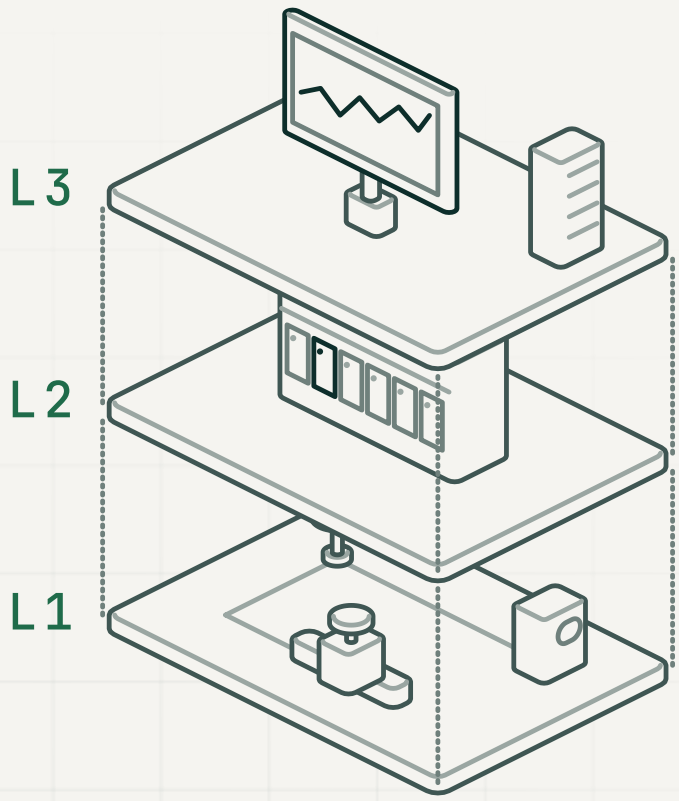
Spare point capacity: at least 15–20%.

L1 Field

Sensors, actuators, valves, meters.

CO_2 sensors drift out of calibration in 2–3 years.

FIG. 02 · LEVELS



• CONCEPTS

Five concepts.

The key to any specification.

01 **Data point**
A value with an address. The unit of measure of a BMS project.

02 **Control loop**
Measure, compare with the setpoint, act. Tuning decides whether the plant is stable.

03 **Sequence of operation**
How the plant behaves in every state. The most important document, and the most often missing.

04 **Schedule**
The operating calendar. The cheapest saving in a building.

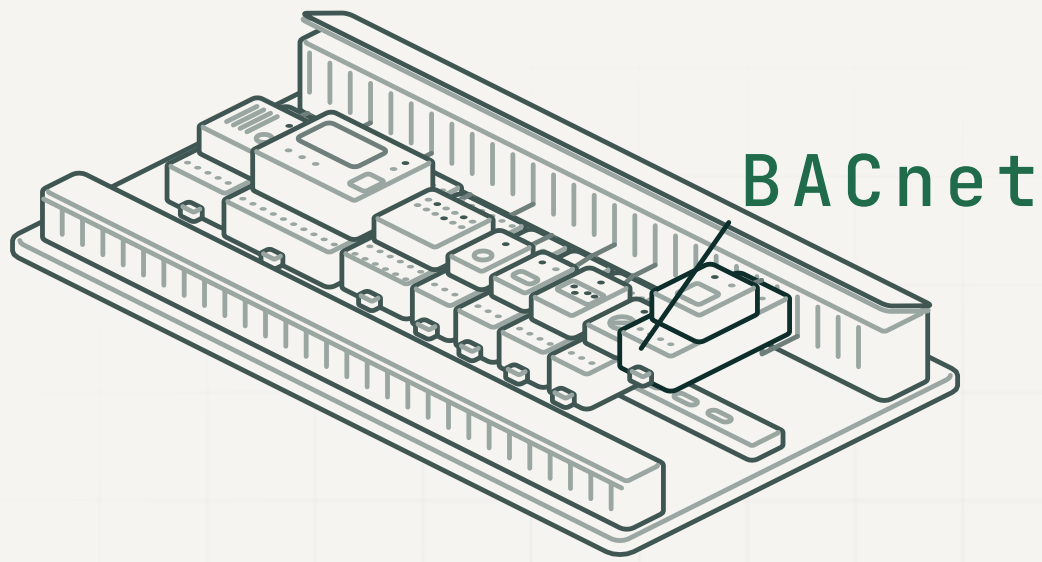
05 **Interlock**
A non-negotiable condition: the fan does not start until the damper is open.

• PROTOCOLS

Six protocols.

A ten-year commercial decision.

FIG. 03 · THE AUTOMATION PANEL



BACnet	The backbone: controllers, AHUs, chillers
Modbus	Third-party equipment: chillers, UPS, power analysers
KNX	Lighting, blinds, room control
DALI	Addressable luminaires
M-Bus	Heat, water and gas meters
LON	Declining; common in buildings fitted out 2000–2015

• OPENNESS

A closed system looks fine in year one.

It shows in year five, when the owner wants 40 more points for a refitted floor and gets a single quote.



• THE MINIMUM TO WRITE INTO THE SPECIFICATION

- Native BACnet/IP at automation level
- Modbus integration for third-party equipment
- Full handover of point maps and the software project
- Licences in the client’s name

• RESULTS

What a BMS can deliver

when properly commissioned.

10–20% less HVAC energy use, estimated	**5–15%** less total energy use, on normalised bills
15–30% fewer unplanned call-outs over two years of disciplined operation	**15 min** trend-log interval per zone: every comfort complaint can be checked
1–3 years typical payback for low-cost optimisation of an existing system	**3–6 years** payback for a capital BACS upgrade

Estimates from delivered projects, market ranges and literature; the energy figures assume a building without coordinated automation. Results depend on the starting point.

• COST

What it costs.

And what budgets leave out.

9–18 EUR/m² class A offices	**5–10** EUR/m² class B offices
90–320 EUR per data point, the alternative benchmark	**4–12%** of the investment a year for maintenance: 4–7% basic, 7–12% extended
8–18% of the software value a year in licence fees	**LEFT OUT OF BUDGETS** Supervision licences, third-party integration, commissioning, as-built documentation.

Excluding VAT. Indicative, not a quote. The EUR/m² ranges cover equipment, installation and commissioning.

• MISTAKES

Eight mistakes that keep coming back.

01 A specification copied from another project	02 A system sized to the last point, with no spare
03 THE MOST EXPENSIVE **Commissioning cut from the budget**	04 Poorly configured alarms: at 400 a day, all ignored within two weeks
05 Meters installed without communication	06 A closed system, accepted without anyone noticing
07 Missing or token as-built documentation	08 Nobody in the organisation responsible for the system

A system installed but never tuned uses as much as no system at all. And one tuned in August is not tuned for January.

• BEFORE SIGNING

Ten questions

for the integrator, answered in writing.

01 Which protocols are native, and what goes through a gateway?

02 Who owns the licences, and what do 200 more points cost in year five?

03 Is the software project handed over editable, with backups?

04 Who writes the sequences of operation, and when can they be seen?

05 How many commissioning hours are included, and what is tested?

06 How much spare point capacity does the design have?

07 Who answers at 2 a.m., how fast, and what does the contract say?

08 Three similar buildings commissioned by the proposed team, not just the company?

09 What happens if the supplier disappears?

10 Who has remote access, and how?

Save this list. Under 15% spare points: paid changes from year one.

• BY ROLE

What it means for each role.

01 Owner and investor

Lower regulatory risk, a protected asset value. First internal question: what rated output is installed across the portfolio?

02 Facility manager

The quick win is running it properly: real schedules, cleaned-up alarms, retuned loops, failed points fixed.

03 Technical director

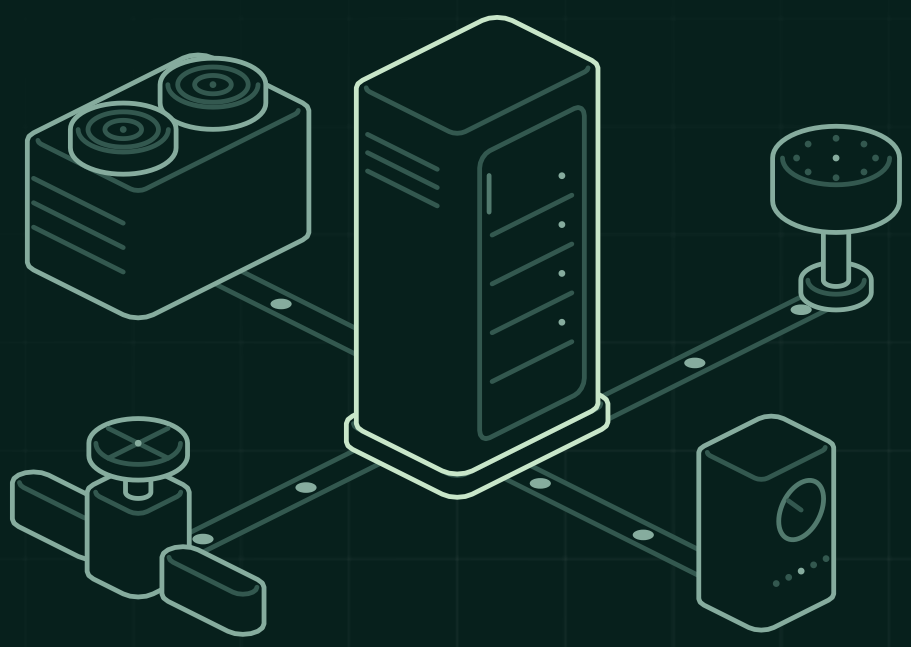
The win is in the specification: native open protocols, the software project handed over, licences in the client’s name, with the expansion price fixed upfront.

Send this to the colleague in the right role.

You don’t buy a BMS. You specify it.

The buildings that run well are the ones where the point list and the sequences of operation were written before the project went out to tender.

FIG. 04 · ALL INTO THE BMS



• THE FULL GUIDE (IN ROMANIAN)

sovitech.ro/ghid/sisteme-bms-cladiri

The editable BMS specification template (point list, sequences, protocol and licensing clauses, acceptance criteria) is available on request at sovitech.ro/contact.